Name: _____

Radioactivity 1 H

Date:

Time:

Total marks available: 50

Total marks achieved: _____

Q1.

Carbon-14 is a radioactive substance that emits beta minus (β^-) particles.

(i) Which of these is the unit for the activity of a sample of carbon-14?

(1)

- ☐ **A** Pa
- ☐ **B** Nm
- ☐ **C** Gs
- ☐ **D** Bq

(ii) Which row of the table gives a correct comparison of the penetrating abilities of alpha, beta and gamma radiation?

(1)

	least penetrating ⟶ most penetrating
☐ **A**	alpha, beta, gamma
☐ **B**	alpha, gamma, beta
☐ **C**	gamma, beta, alpha
☐ **D**	gamma, alpha, beta

(iii) The mass of a beta minus particle is 9.1×10^{-31} kg.

Calculate the kinetic energy of a beta minus particle moving with a speed of 3.8×10^{6} m/s.

(2)

kinetic energy = ... J

(Total for question = 4 marks)

Q2.

(a) Positrons are antiparticles of electrons.
The photograph shows a PET scanner that uses positrons to locate the site of a brain tumour.

Positrons are emitted by the radioactive isotope fluorine-18.
The fluorine-18 is attached to glucose.
The glucose is injected into the patient's blood stream and collects at the site of the tumour.

(i) Fluorine-18 has a half-life of 1.8 hours.
State one advantage and one disadvantage of using a substance with such a short half-life.

1. Advantage

(1)

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2. Disadvantage

(1)

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(ii) When a positron meets an electron they annihilate each other.
Explain how this enables the site of the tumour to be located.

(4)

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*(b) Positrons (β+ particles) are emitted from the nuclei of some atoms.
Electrons (β− particles) are emitted from the nuclei of other atoms.
The table gives some information about quarks.

quark	charge (compared to the charge on a proton)
u	+2/3
d	−1/3

Describe, in terms of quarks, how β+ particles are emitted from the nuclei of some atoms and β− particles are emitted from the nuclei of others.

(6)

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(Total for Question = 12 marks)

Q3.

Some rocks contained uranium when they were formed.

Radioactive decay in these rocks produces radon gas.

Explain why people living near these rocks have an increased health risk from background radiation.

(3)

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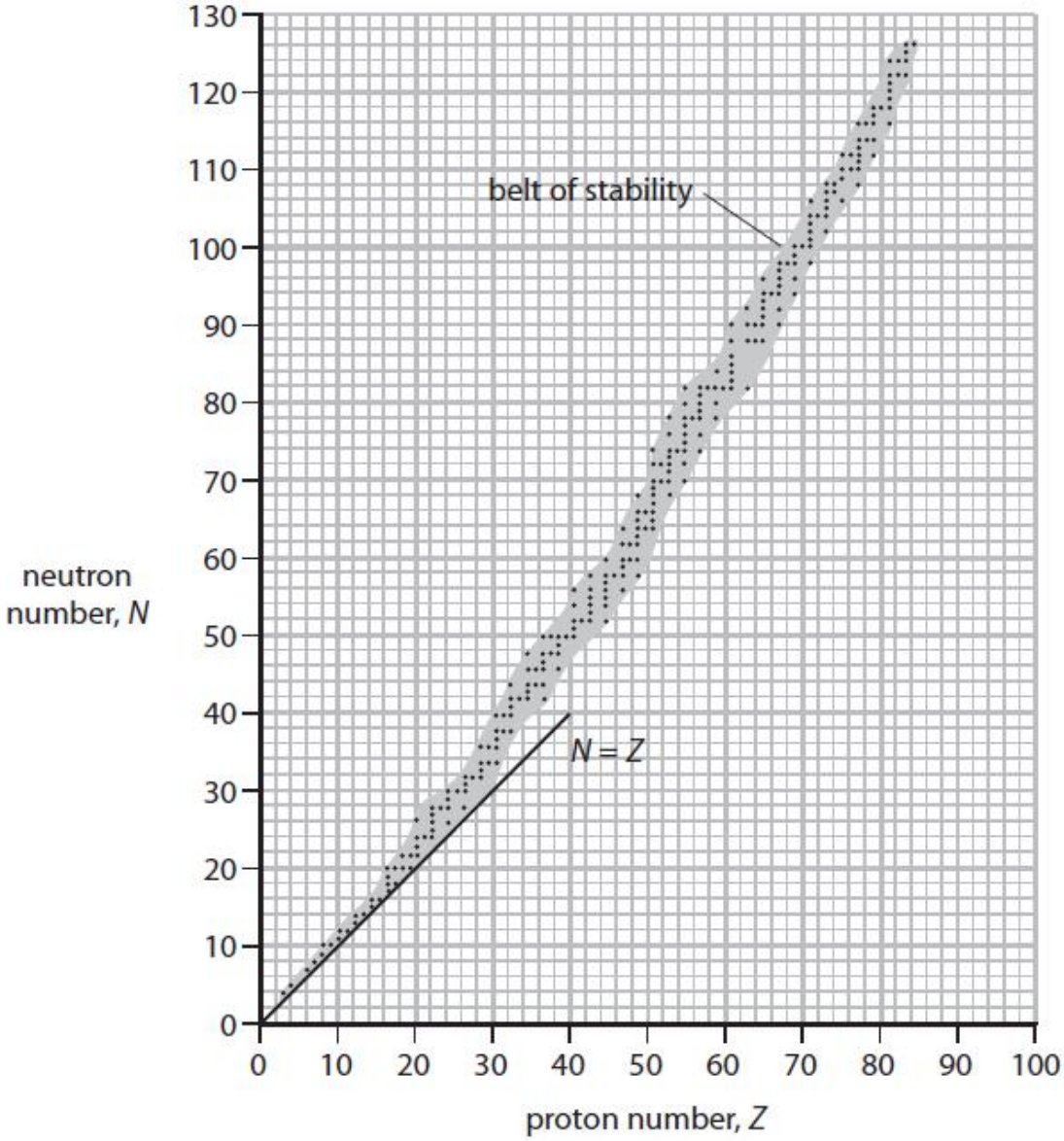
Q4.

(a) The diagram shows the stability curve for nuclear isotopes.



Complete the sentence by putting a cross (☒) in the box next to your answer.
An isotope above the curve will undergo β^- decay because it has

(1)

☐ **A** too few protons
☐ **B** too many protons
☐ **C** too few neutrons
☐ **D** too many neutrons

(b) Which statement is correct for β^+ and β^- particles?

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** a β^+ is positively charged and a β^- is negatively charged
☐ **B** the mass of a β^+ is 1800 times the mass of a β^-
☐ **C** the charge on a β^+ is twice the charge on a β^-
☐ **D** a β^+ is a proton and a β^- is an electron

(c) Carbon-14 decays by emitting a β^- particle to form an isotope of nitrogen.

Complete the nuclear equation for this decay by filling in the boxes.

(2)

$$^{14}_{6}C \rightarrow \square\,N + \square\,\beta^-$$

(d) Protons and neutrons both contain quarks.

Describe the arrangement of quarks in a proton.

(2)

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(e) Explain what happens to a nucleus during β^+ decay.

(2)

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Q5.

A G-M tube is connected to a counter.
A teacher places the G-M tube near to a radioactive source.
A student starts the counter and clock at the same time and writes down the readings shown on the counter every 15 s.

The student plots the readings with a line of best fit, as shown in Figure 10.

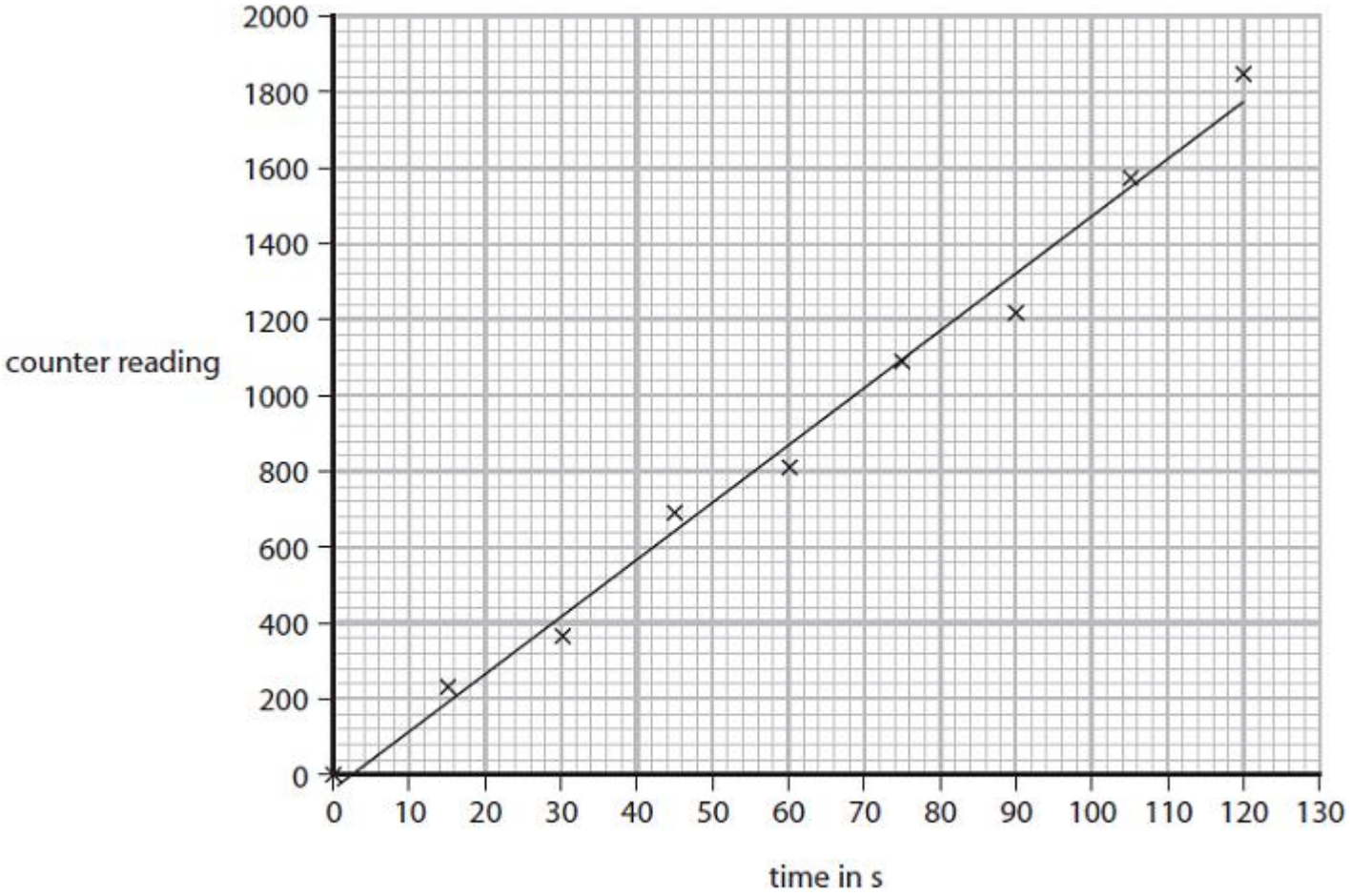

Figure 10

(i) Calculate the average count rate, in counts / s, from the graph.

Show your working on the graph.

(2)

average count rate = ... counts / s

(ii) The student says that the experiment must have been done carelessly because the data seemed quite scattered
away from the best fit line.

The teacher claims such results should be expected in radioactivity experiments.
Justify the teacher's claim.

(2)

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(Total for question = 4 marks)

Q6.

This question is about radioactivity.

Radium-223 is a radioactive substance.

Radium-223 is an alpha emitter.

The half-life of radium-223 is 11 days.

A radioactive source contains 1.7×10^{23} nuclei of radium-223.

Calculate the number of radium-223 nuclei remaining in the source after a time of 33 days.

(2)

number of radium-223 nuclei remaining = ...

(Total for question = 2 marks)

Q7.

The isotope fluorine-18 is used in PET scans, which help to detect cancers.

Figure 7 shows a radioactive decay curve for a sample of fluorine-18.

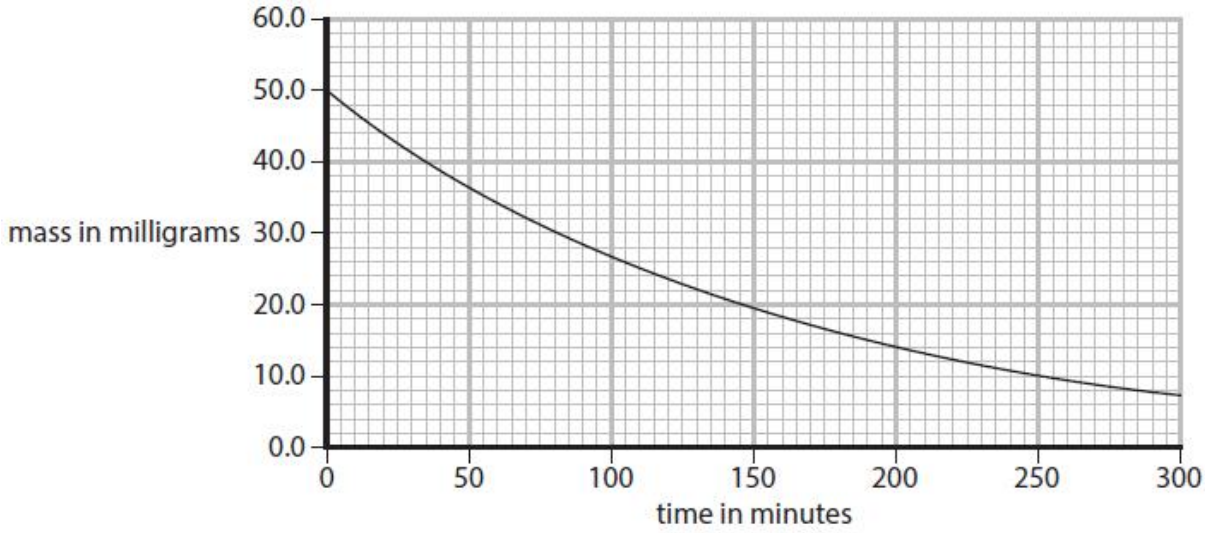

Figure 7

(i) Use the graph to calculate the percentage of fluorine-18 remaining after 150 minutes.

(2)

percentage of fluorine-18 remaining after 150 minutes = ... %

(ii) Fluorine-18 releases positrons.

Describe what happens when positrons meet electrons.

(2)

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(Total for question = 4 marks)

Q8.

In the Sun, four protons start the process of nuclear fusion.

These protons combine and finally produce a helium nucleus.

The helium nucleus has a smaller mass than the four protons.

This difference in mass is converted to energy.

Four protons have a total mass of 6.69×10^{-27} kg.

A helium nucleus has a mass of 6.64×10^{-27} kg.

Calculate the percentage of the original mass that has been converted to energy.

(3)

percentage of mass converted to energy ..... %

(Total for question = 3 marks)

Q9.

In 2023, Finland produced 4.4 GW of power from nuclear power plants.

The UK produced 8.9 GW of power from nuclear power plants.

Calculate the ratio of nuclear power produced in Finland compared with the UK as

$$\frac{\text{power production in Finland}}{\text{power production in the UK}}$$

Give your answer to 2 significant figures (2 s.f.).

(2)

ratio = ...

(Total for question = 2 marks)

Q10.

The intensity of gamma radiation can be measured using a Geiger-Müller tube and counter.

The count rate recorded by the counter tube depends on how far away the Geiger-Müller tube is from the gamma radiation source.

The equation relating count rate to distance from the source is

$$\text{count rate} = \frac{k}{d^2}$$

where d is the distance from the source and k is a constant.

A Geiger-Müller tube is placed 0.70 m from a source of gamma radiation.
The counter displays a count rate of 85 000 count per minute.

Calculate the count rate recorded when the Geiger-Müller tube is placed 1.3 m away from the same gamma radiation source.

(3)

count rate = ... counts per minute

(Total for question = 3 marks)

Q11.
Radioactive tracers can be used when scanning a person's kidneys.

A radioactive isotope is injected into a person's blood stream.

The isotope emits radiation.

As the blood flows through the kidneys, this radiation is detected outside the body by a scanner.

(i) What type of radiation travels from the kidney to the scanner?

(1)

- ☒ **A** alpha
- ☒ **B** beta plus
- ☒ **C** beta minus
- ☒ **D** gamma

(ii) During the scan, a technician needs to take readings for about 30 minutes.

The half-life of the isotope used is about 6 hours.

1. State why an isotope with a half-life of about 6 minutes is **not** suitable.

(1)

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.....

2. State why an isotope with a half-life of about 6 days is **not** suitable.

(1)

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(iii) State **two** ways of reducing the radiation risks to the technician.

(2)

1 .....

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2 .....

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(Total for question = 5 marks)